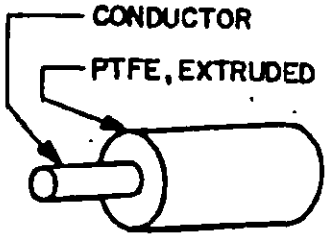
MIL-W-16878/27(NAVY)
10 August 1981
SUPERSEDING
MIL-W-16878/5A(NAVY) (IN PART)
5 July 1961
(See supersession data herein)

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE (PTFE) INSULATED, 260°C, 1000 VOLTS, EXTRUDED INSULATION

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878.



SH 11875

FIGURE 1. Construction.

TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material 2/ 3/	Coating		Diameter (inch)	
					Min	Max
32	1 X 32	Copper	Nickel	0.0080	0.034	0.042
32	1 X 32	H.S.C.A.	Nickel	.0080	.034	.042
32	1 X 32	C.C. Steel	Nickel	.0080	.034	.042

See footnotes at end of table.

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TABLE I. Construction details. - Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material[2/3/]	Coating		Diameter (inch)	
					Min	Max
32	7 X 40	Copper	Nickel	0.010	0.036	0.044
32	7 X 40	H.S.C.A.	Nickel	.010	.036	.044
30	1 X 30	Copper	Nickel	.0100	.036	.044
30	1 X 30	H.S.C.A.	Nickel	.0100	.036	.044
30	1 X 30	C.C. Steel	Nickel	.0100	.036	.044
30	7 X 38	Copper	Nickel	.012	.038	.046
30	7 X 38	H.S.C.A.	Nickel	.012	.038	.046
28	1 X 28	Copper	Nickel	.0126	.039	.047
28	1 X 28	H.S.C.A.	Nickel	.0126	.039	.047
28	1 X 28	C.C. Steel	Nickel	.0126	.039	.047
28	7 X 36	Copper	Nickel	.015	.041	.049
28	7 X 36	H.S.C.A.	Nickel	.015	.041	.049
26	1 X 26	Copper	Nickel	.0159	.042	.050
26	1 X 26	H.S.C.A.	Nickel	.0159	.042	.050
26	1 X 26	C.C. Steel	Nickel	.0159	.042	.050
26	7 X 34	Copper	Nickel	.019	.045	.053
26	7 X 34	H.S.C.A.	Nickel	.019	.045	.053
26	19 X 38	Copper	Nickel	.020	.045	.053
26	19 X 38	H.S.C.A.	Nickel	.020	.045	.053
24	1 X 24	Copper	Nickel	.0201	.046	.054
24	1 X 24	H.S.C.A.	Nickel	.0201	.046	.054
1/24	1 X 24	C.C. Steel	Nickel	.0201	.046	.054
24	7 X 32	Copper	Nickel	.024	.050	.058
24	7 X 32	H.S.C.A.	Nickel	.024	.050	.058
24	19 X 36	Copper	Nickel	.025	.050	.058
24	19 X 36	H.S.C.A	Nickel	.025	.050	.058
22	1 X 22	Copper	Nickel	.0254	.051	.060
22	1 X 22	H.S.C.A.	Nickel	.0254	.051	.060
1/22	1 X 22	C.C. Steel	Nickel	.0254	.051	.060
22	7 X 30	Copper	Nickel	.030	.056	.064
22	7 X 30	H.S.C.A.	Nickel	.030	.056	.064
22	19 X 34	Copper	Nickel	.032	.056	.064
22	19 X 34	H.S.C.A.	Nickel	.032	.056	.064
20	1 X 20	Copper	Nickel	.0320	.058	.066
20	1 X 20	H.S.C.A.	Nickel	.0320	.058	.066
20	7 X 28	Copper	Nickel	.038	.064	.072
20	7 X 28	H.S.C.A.	Nickel	.038	.064	.072
20	19 X 32	Copper	Nickel	.040	.064	.072
20	19 X 32	H.S.C.A.	Nickel	.040	.064	.072

See footnotes at end of table.

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TABLE I. Construction details. - Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
					Diameter (inch)	
		Material 2/ 3/	Coating		Min	Max
18	1 X 18	Copper	Nickel	0.0403	0.066	0.076
18	7 X 26	Copper	Nickel	.048	.074	.084
18	19 X 30	Copper	Nickel	.050	.074	.084
4/ 16	1 X 16	Copper	Nickel	.0508	.077	.089
4/ 16	19 X 29	Copper	Nickel	.057	.083	.095
14	19 X 27	Copper	Nickel	.072	.098	.114
12	19 X 25	Copper	Nickel	.091	.117	.133
12	37 X 28	Copper	Nickel	.089	.115	.131
10	37 X 26	Copper	Nickel	.111	.137	.153
10	105 X 30	Copper	Nickel	.120	.146	.162
8	133 X 29	Copper	Nickel	.169	.199	.219
6	133 X 27	Copper	Nickel	.213	.253	.273

1/ Inactive for new design.
2/ H.S.C.A. stands for high strength copper alloy.
3/ C.C. Steel stands for copper clad steel.
4/ Under MPD-1506, the 16 AWG size was heavier.

REQUIREMENTS:

Visual and mechanical inspection: Applicable.
Spark test: 5.0 kilovolts (kV).
Impulse dielectric test: 8.0 kV.
Tank test: 2.0 kV.
Dielectric withstanding voltage: 2.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20 °C
K = 50,000
D = Maximum average diameter of finished wire
d = Conductor diameter

Cold bend: Condition 4 hours at minus 65°C ± 1°C (see table II).

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TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter) (inches)
32 through 16	1
14, 12	2
10, 8, 6	3

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Applicable.
Heat resistance: Condition at 290°C.
Flammability: Not applicable.
Soldering: Applicable.
Heat aging: Not applicable.
Insulation tensile strength: 3000 pounds force per square inch (minimum).
Insulation elongation: 150 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire: See table I of MIL-W-16878.

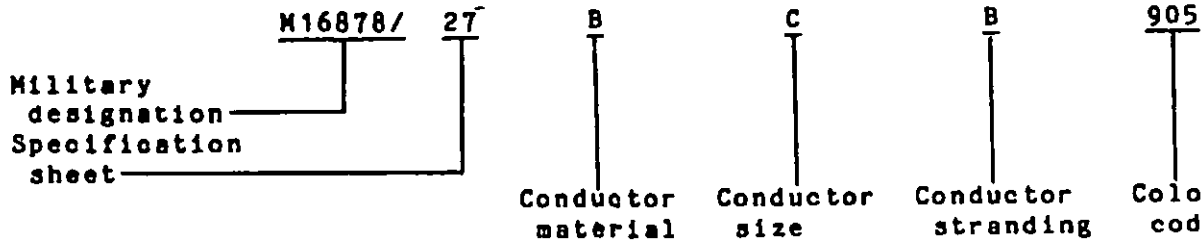
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TABLE III. Put up tabulation.[1/]

AWG size	Percentage of order	Length (ft) one continuous length	Maximum no. of lengths (per spool or reel)	
			500 ft	1000 ft
32 to 22	25 25 50	Under 75 75 - 300 301 or more	3	4
20 to 16	Unacceptable 20 50 30	Under 75 75 - 100 101 - 300 301 or more	4	5
14 to 10	Unacceptable 40 50 10	Under 50 50 - 100 101 - 300 301 or more	5	6
8 and larger	10 70 20	Under 50 50 - 100 100 or more	6	7

[1/] Each spool or reel shall be marked accordingly with the length in feet and location of each piece.

Military part number (see MIL-W-16878):



Supersession data: The wire covered by this specification sheet supersedes MIL-W-16878/5A(NAVY), type EE, 260°C rated, with extruded insulation.

Review activities:
Navy - AS, EC
User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N300-26)

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